Exercise: 20

Cell cycle

Aim: To study and identify the stages of Cell cycle.
Principle: A series of events leading to the formation of new cell is known as cell cycle. The phenomenonal changes leading to formation of new population take place in the cell cycle. The series of events include, G1, S, G2, and M.
Requirements: Model / Photograph / picture of Cell cycle.

Diagnostic features:

1. G_1 Phase:

- G_1 or first gap phase, cells are metabolically active and produces proteins, lipids, carbohydrates, cell organelles and RNA.
- A non-dividing cell does not proceed beyond G 1 phase.

2. S Phase:

- S Phase or synthesis phase duplication of DNA takes place.
- The centrioles duplicate in the cytoplasm.
- DNA content increases from 2 C to 4 C.

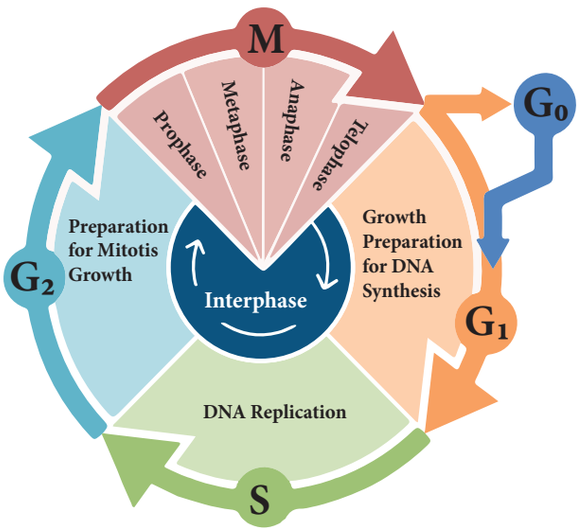


Figure 20: Cell cycle

3. G_2 Phase:

- G_2 Phase or second gap phase cell grow continuously by protein and cell organelle synthesis, mitochondria and chloroplast divide and DNA remain in 4 C.
- Tubulin, microtubules and Maturation Promoting Factor are formed.
- Spindle begins to form and nuclear divisions follows.

4. M Phase:

- M Phase or Mitotic Phase is short, in this stage a single cell divided into two daughter cells.
- The number of chromosomes in the parent and the daughter cells remain the same so it is also called as equational division.
- This phase is divided into Prophase, Metaphase, Anaphase and Telophase.

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